RESEARCH STRATEGY

SIGNIFICANCE

Early Identification Catalyzes Early Intervention, *Especially for Vulnerable Infants*

Autism spectrum disorder (ASD) is a neurodevelopmental disorder that affects ~2.3% of the population and carries a lifelong financial cost of $1.4-2.4 million per individual, thus representing a pressing public health challenge.[30,31] Early intervention plays a pivotal role in the lives of children and families affected by ASD – improving functional outcomes for the individual, enhancing the quality of life for the family, and reducing the lifelong financial cost.[32,33] Unfortunately, with an ASD diagnosis and intervention onset occurring after age 4-6 years,[34,35] we are missing out on a critical period for early intervention that could capitalize on peak neuroplasticity in the first 2 years of life.[36,37] Later identification of ASD occurs disproportionately in more vulnerable populations,[30,31,38] such as those with medical comorbidities, neurodevelopmental impairments (NDIs),[32] as well as those from marginalized racial/ethnic groups and lower socioeconomic status (SES), all of which are highly prevalent in infants born preterm (PT).

Across the world, 15 million PT infants are born each year.[39] Prematurity is *the* leading cause of short and long-term morbidities[40,41] and PT infants are at a significantly increased likelihood for ASD and NDI. The risk for NDI outcomes increases exponentially with lower gestational age (GA)[15] and the occurrence of neonatal morbidity in the Neonatal Intensive Care Unit (NICU).[21,22,42] Socioeconomic disparities play a substantial role in PT birth outcomes, with evidence for multiplicative effects of SES and GA on NDI outcomes.[26,43,44] The burden of PT birth equates to at least $26 billion in the United States (US), which includes medical, educational, and lost productivity expenses.[45] The American Academy of Pediatrics emphasizes the importance of early intervention and coordinated efforts for follow-up care to maximize childhood health and development outcomes.[46]

The rate of ASD in PT infants is more than triple that of the general population,[3] with recent reports of a 10-fold increase for very PT infants (VPT; born <32 weeks GA)[4] and a 20-fold increase for extremely PT infants (<28 weeks GA).[5] Thus, PT infants are at an elevated likelihood for ASD while also experiencing medical, developmental, and socioeconomic risk factors that put them at a very high risk of *later identification* of ASD and *even later* intervention. **This is a critical barrier in the field** – earlier identification of ASD within the first year of life of PT infants can help address this health disparity by *catalyzing earlier intervention,* leading to greater *improvements in long-term outcomes*. Evidence that forms the **scientific basis of this study** includes: 1) the importance of autonomic nervous system (ANS) in environmental adaptation, social interaction, and learning, 2) the presence of ANS dysfunction at birth in VPT infants and throughout the lifespan of individuals with ASD, 3) the exacerbation of ANS dysfunction by PT morbidities, and 4) the elevated prevalence of ASD and NDI phenotypes among VPT infants. This strong scientific foundation and the absence of clear mechanistic explanations for distinct developmental pathways present three gaps:

(1) **Multimodal measurement** has not yet been leveraged to analyze the specific and combined contributions of ANS subcomponents (i.e., sympathetic and parasympathetic) to PT autonomic dysfunction and developmental outcomes.

(2) **Prematurity-associated morbidities** have a compounding effect on ANS dysfunction, but their impact on the association between ANS and developmental outcomes is not well understood.

(3) **Prospective, comprehensive characterization of trajectories** of ASD-specific features and neurodevelopmental impairment throughout infancy in VPTs is lacking.

These gaps **call for further research** on ANS dysfunction as a mechanism for ASD in VPT infants.

ANS Dysfunction: A Mechanistic Hypothesis for Increased Incidence of ASD in PT Children

Existing evidence suggests that ANS dysfunction is experienced by both PT infants and children with ASD. In this study, we will evaluate **ANS dysfunction** as a novel mechanism for the increased incidence of ASD in VPT infants by surveilling NICU-based indicators of ANS functioning and comprehensively phenotyping the emergence of ASD symptoms and NDI across the first 3 years of life. Determining the association between PT ANS dysfunction and ASD/NDI outcomes by age 3 years will significantly improve scientific knowledge of ASD pathogenesis. Ultimately, it will pave the way for translatable neonatal indicators of risk that will lead directly to tailored developmental follow-up and **reduce disparities in timely diagnosis and intervention** for this vulnerable population. This study is well aligned to address PAR-21-201 which calls for studies designed to elucidate the etiology, epidemiology, and diagnosis in relation to ASD.

Significant ANS maturation occurs in the second half of pregnancy, and PT birth (born <37w GA) significantly disrupts ANS development and the overall functioning of the PT newborn.[47,48] For PTs, reduced ANS activity fails to normalize at term age equivalent,[49] and may alter long-term temporal programming of ANS maturation.[50] Following birth, ANS plays a vital role in the adaptive competence of the newborn through balanced adaptive responses of two major subcomponents: the sympathetic nervous system (“fight or flight”; SNS) and the parasympathetic nervous system ("rest and digest"; PNS). In response to ever-changing environmental contexts, these systems enact physiological changes that regulate basic functions, such as heart rate and body temperature. In early prenatal development, physiological functions are driven primarily by sympathetic influence. Higher cortical processes begin to integrate autonomic control in the late prenatal and early postnatal period leading to increasing influence of the PNS. As ANS matures, parasympathetic tone serves as the foundation for several higher-level behaviors that support calm states of alertness and sustained attention, ultimately providing countless opportunities for social and language learning. In this way, ANS is considered a "developmental resource" in the neonatal and early infant period that mediates infant-environment interactions.[51]

In the case of PT birth, a drastic transition to the extrauterine environment occurs amidst ongoing ANS maturation and is a clash that is compounded by exposure to medical stressors. As a result, autonomic maturation is impaired throughout the PT period, initiating a developmental cascade that may disrupt the physiological adaptation to the environment and integration of autonomic control into later-developing, higher-level cortical processes important for social learning and cognitive development. These early disruptions may be at the root of the emergence of ASD symptoms (social-communication delays, social interaction impairments, repetitive behaviors, atypical sensory reactivity). Emerging evidence for this developmental pathway from our group links early infant ANS/attention regulation in full-term infants to later social-communication and ASD.[18,52,53]

1. Multimodal Measurement of PT ANS Dysfunction

Current studies of PT ANS dysfunction are largely limited to unimodal measurement of ANS function, most often heart rate indices (HRIs), such as heart rate variability, derived from electrocardiographic (ECG) recordings[54]. The current study overcomes this limitation by leveraging multimodal measurement of ANS function that includes recordings of **peripheral and central body temperature** and **ECG**.

Temperature-Derived Measures of ANS Dysfunction. Body temperature is regulated in part by vasomotor tone, or constriction and dilation of blood vessels, which may become altered with autonomic instability.[55] In an infant with a healthy, stable ANS, the temperature is higher for the central body than the periphery, with a Central (abdominal) - Peripheral (foot) Temperature difference (CPTd) of 1-2º C.[56] Thus, CPTd is a measure of the **thermal gradient**, or blood perfusion across the infants' body. The SNS plays a critical role in preventing abnormal thermal gradients through peripheral vasoconstriction.[57] Under normal conditions, the vasomotor center in the brain continuously transmit signals to the sympathetic vasoconstrictor nerve fibers throughout the body.[58] This continual firing is called vasoconstrictor tone and is responsible for maintaining vasomotor tone, which allows constriction of peripheral vessels to conserve heat. Peripheral temperature correlates with peripheral blood flow.[59] As blood flow to the extremities increases, higher peripheral temperatures occur. Our research has found that assessing vasomotor tone through surveillance of the CPTd offers a novel approach to evaluating ANS dysfunction.[60,61] Longitudinal abnormal thermal gradients (CPTd <0° or CPTd≥2°C) are a sign of microcirculatory dysfunction and can be used as a sign of ANS dysfunction (likely driven by abnormal SNS activation), which may be an indicator of neurodevelopmental outcome and ASD.[7] As in our previous work,[61,62] this study will use continuous measurement of PT abnormal thermal gradients as an index of ANS dysfunction.

ECG-Derived Measures of ANS Dysfunction. ECG recordings of heart activity yield several **HRIs** that can characterize ANS function, including beat-to-beat heart rate, transient heart rate accelerations/decelerations (i.e., tachycardias and bradycardias), and heart rate variability (HRV), and hear rate variability at the frequency of respiration (i.e., respiratory sinus arrhythmia; RSA). Each HRI represents unique and combined effects of sympathetic and parasympathetic activity, where sympathetic activity generally increases heart rate and parasympathetic activity reduces heart rate. The magnitude of HRV and RSA measures increase over the first years of life, and higher magnitudes indicate a more adaptive, balanced ANS system.[63] This process is disrupted in PT infants, resulting in an increase in RSA magnitude that is 2-3 times slower than that of term infants by 6 months age.[64] Time and frequency domain analyses of HRV show that ANS function is adversely affected by PT-related morbidities such as mechanical ventilation, infections, and patent ductus arteriosus (PDA).[20] This detrimental influence of PT birth on RSA and HRV has been found to persist into childhood and even adulthood.[65],[66] Existing research, including our own, link abnormal HRIs (e.g., transient heart rate accelerations) in the PR period with the later emergence of ASD, with atypical RSA emerging in later infancy (age 2-3 years) for those with ASD, suggesting a potential decline in parasympathetic regulatory mechanisms.[7,67] While RSA could be affected by external influences on respiration (e.g., ventilatory rate, pressure support, and continuous positive airway pressure), previous studies have documented success in measuring RSA in PT infants on mechanical ventilation and we will include respiratory support and severity of lung disease as covariates in all analyses of ANS dysfunction based on RSA.[68,69] HRIs can be obtained through standard cardiopulmonary monitoring systems and, for this study, will be obtained via HeRO bedside monitors,[70] as we have done previously.[7,62] Continuous heart activity recording offers a novel way to evaluate ANS function, including both sympathetic and parasympathetic activation and balance, in VPT infants in the NICU.

The multiplicative effects of ANS dysfunction on several biological functions make mechanistic interpretations of unimodal data insufficient. For example, abnormal thermal gradients may be caused by vasoconstriction resulting from sympathetic overactivation or by reduced vasodilation caused by parasympathetic under activation. Imbalance of the SNS and PNS has been suggested as an ASD-specific biomarker,[10] but a debate remains on what precisely causes this imbalance (e.g., overactive sympathetic or underactive parasympathetic). Multimodal assessment (temperature, ECG) that includes **multiple** biological indicators will allow for a clear, specific description of ANS function and a nuanced understanding of adaptive and maladaptive sympathetic and parasympathetic responses. This will help parse contributions of SNS and PNS in the overall evaluation of ANS dysfunction and test their relative predictive contributions to ASD and NDI.

2. Prematurity-Associated Morbidities, ANS Dysfunction, and Neurodevelopmental Outcome

Infants born PT experience multiple morbidities (see Table 1) and higher major morbidity rates are associated with worse neurodevelopmental outcomes.[71]

Table 1. Preterm Infants <32 weeks GA Morbidities and Incidence

Morbidity	Definition	Incidence
Severe Brain Hemorrhage	Grade III-IV, Periventricular Leukomalacia	6-19%
Sepsis	Late onset blood infection > 7 days of age	9-50%
Necrotizing Enterocolitis (NEC)	Infection of the GI tract, requiring medical and/or surgical treatment	5-15%
Bronchopulmonary Dysplasia (BPD)	Chronic Lung Disease, requiring oxygen use at 36 weeks postnatal age	32-44%
Patent Ductus Arteriosus (PDA)	The Ductus Arteriosus fails to close post birth and requires medical or surgical treatment for closure if hemodynamically significant	5-60%
Retinopathy of Prematurity (ROP)	Disorder of the developing retina of preterm infants that can lead to blindness	3-42%

Preterm Morbidities and ANS Dysfunction. The risk for morbidities increases with earlier GA[72,73] and recent research suggests that PT morbidities may mediate the adverse effect of GA on PT ANS dysfunction.[20,71,74] All PT morbidities are associated with periods of ANS dysfunction due to the morbidity-related stress on infants' already fragile homeostasis – many directly attacking the brain, heart, or lungs. Some researchers posit that ANS dysfunction underlies the need for oxygen in PTs and, in turn, this increased oxygen exposure can cause hyperoxia and induce oxygen radical disease, which links ANS dysfunction to BPD, ROP and NEC morbidities in VPT infants.[72,75] Thus, early-life ANS dysfunction may both cause and exacerbate PT morbidities.

Preterm Morbidities and Developmental Outcome. As ANS plays a large regulatory role in mitigating stressors of extrauterine life in the VPT infant, the combined effects of ANS dysfunction and PT morbidities can greatly decrease optimal health outcomes. The risk for ASD and NDI increases with decreasing GA and limited research has shown an association between PT morbidities and ASD.[76] For example, the occurrence of 2-4 PT-related morbidities has been associated with positive ASD screening (M-CHAT-R/F)[24] and PT sepsis has been associated with a later ASD diagnosis.[77] Some medical treatments for morbidities (e.g., steroids for BPD) have also been associated with ASD.[78] This already limited research largely comes from retrospective analysis of medical records and has not addressed how PT morbidities *moderate* associations between ANS dysfunction and *dimensional* ASD *and* NDI outcomes. Aim 2 will evaluate these PT morbidities as potential moderators of the association between ANS dysfunction in the PT period and ASD/NDI outcomes at 36 months.

Social Drivers of Health (**SDOH**). It is also critical to consider the role of **SDOH** in child development and mental health research.[79] Demographic factors and SDOH, such as race/ethnicity, maternal education level, insurance, and housing stability, have major effects on access to maternal prenatal and postnatal healthcare as well as infant and child healthcare, developmental supports, and medical and mental health diagnoses.[80,81] These barriers to care can heavily influence preterm morbidities and neurodevelopmental outcomes, as our work has suggested.[82-86] Likewise, early infant experience, including maternal stress, postnatal hospitalizations, and early interventions, can influence outcomes. As such, demographic variables, SDOH,[87,88] maternal stress,[89] and infant context (intervention, hospitalizations) will be assessed and accounted for in analytic models.

3. Comprehensive Characterization of Emerging ASD-Specific Symptoms and NDI

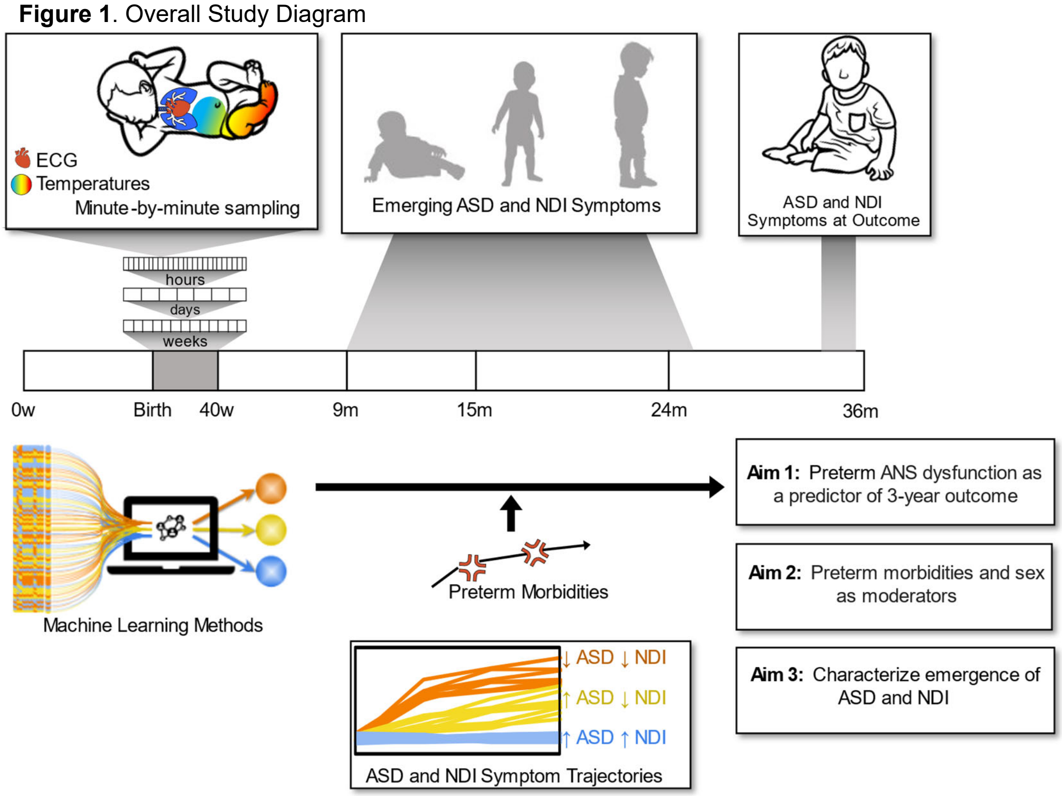
Rates of both ASD and NDI in VPT infants are significantly greater than that of the general population,[3] with many overlapping symptoms common to VPTs both with and without ASD including: motor dysfunction, language deficits, social impairment, atypical sensory processing, and executive function challenges.[26] This is further complicated by health disparities wherein PT birth disproportionately effects minoritized and lower SES families and, in turn, lower SES adversely effects developmental outcomes for individuals with and without ASD,[90,91] as our work has also shown.[83] These commonalities complicate differential diagnosis of ASD and NDI. Yet another complication is characterizing development and change during a critical period of rapid brain development and skill acquisition. Our preliminary data, along with existing research,[27,92] suggest differential trajectories of development across domains for VPT infants where strengths and weaknesses are not stable. Moreover, direct observation of ASD-specific symptoms in the first year of life using validated assessment tools[29] have yet to be applied to a cohort of VPT infants. Understanding individual differences and shifts in phenotypic profiles of both ASD-specific characteristics and NDI (cognitive, motor, language impairments) from as early as 12 months of age, over the first 3 years of life, is critical for informed and specific diagnostic and intervention.

INNOVATION

This proposal seeks to shift current research paradigms related to PT birth and ASD by prospectively studying a large cohort of VPT infants for ANS dysfunction during their entire NICU hospitalization and monitoring them for the emergence of ASD/NDI symptoms over their first 3 years of life. Through this investigation, we aim to identify abnormal thermal gradients and HRIs that can be used as a biomarker in infancy to forecast atypical neurodevelopmental outcomes and, specifically, ASD. Study results will inform the novel mechanistic hypothesis that ANS dysfunction is an underlying link between PT birth and ASD outcome. Although our extensive experimental data collection protocol is not meant for direct translation into a revised standard of care, it will inform the relevance of considering ANS monitoring, potentially via contactless, more scalable strategies.

Continuous Measurement of ANS Dysfunction in the NICU. PT infants are born with an exceptionally immature autonomic regulatory system that may not fully develop until well after birth.[65,93-95] We propose to measure ANS dysfunction through continuous recording of biological measures (temperature, ECG) from birth to discharge. Sampling for an average of 9 weeks per infant (average NICU stay = 9 weeks) will yield an average

Figure 1. Overall Study Diagram



of 90,720 minute-by-minute temperature samples and 1512 hours of continuous recordings of ECG *per infant*. This innovative approach will generate an unprecedented volume of longitudinal, multimodal indicators of ANS.

Machine Learning Approach. A machine learning (ML) approach is necessary for the high volume of biological signals that will be collected and processed throughout this study. Our analytic approach leverages ML and artificial intelligence (AI) techniques that support nonlinearity and investigation of multiple ANS measures, while also retaining interpretability, clinical significance, and clinical relevance, of ML algorithm results. Specifically, we will use support vector machine regression, a supervised ML tool for nonlinear regression, in addition to permutation feature importance techniques developed in the context of explainable AI, to identify the most meaningful ANS features in our prediction models. We will also use DBSCAN, an unsupervised ML algorithm to identify clinically meaningful longitudinal phenotypic profiles. These innovative methods represent powerful tools that provide this study with enormous potential for understanding the impact of PT birth on the emergence of ASD symptoms.

APPROACH

Preliminary Data

Thermal Gradients as a Measure of ANS Dysfunction. Our series of studies have shown that ANS dysfunction in VPT infants is observable via abnormal thermal gradients, defined by a condition of warmer, better perfused extremities with simultaneous colder and less perfused core body parts intermittently over their first month of life.[60,61,96-98] CPTd is a measure of microcirculatory dysfunction that can capture this phenomenon.[99] We, and others, have shown that abnormal thermal gradients, defined as a CPTd < 0º or CPTd > 2º C, is correlated with the occurrence of infection in VPT infants. [61,99,100] The state of circulatory dysfunction and centralization of blood flow in relationship to the presence of bacteria, will cause the CPTd to become increased with researchers measuring CPTd values on average of 2º C, but as high as 9º C[99,101] in PT infants with sepsis. In severely ill infants, with decreased GA and immature ANS, the CPTd can be quite small (0-1º C) or negative, indicating warmer periphery than core (see **Fig. 2**).[101,102] We used Grant dataloggers and thermistors in our study (R01NR017872) to measure CPTd to predict infection,[62] a preliminary analysis showed that when an infant had CPTd >2ºC for more than > 50% of the time, it had an 86% higher odds of infection compared to infants without temperature differentials >2ºC.[103] These data demonstrate that continuous CPTd, derived from readily measurable dual temperature vital signs, can be continuously recorded in VPT infants over their NICU hospitalization and effectively utilized as one of two multimodal measures of ANS function.

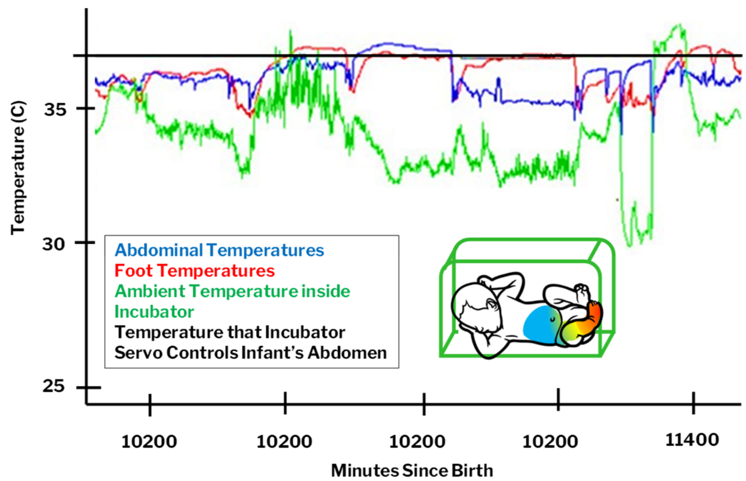


Figure 2. VPT infant body temperatures with intermittent hypothermia, negative central-peripheral temperature difference and normal thermal gradient over 12 hours.

Heart Rate Indices (HRI): A Gold Standard Measure of ANS Dysfunction in the NICU. Heart rate is regulated by the ANS, with sympathetic activation causing accelerations and parasympathetic activation causing decelerations.[1] ANS dysfunction can cause arrhythmias, vascular irritability, and hypo and hypertension.[1] HRIs, including beat-to-beat heart rate, HRV, mean, standard deviation, and skewness of IBI can be recorded through bedside cardio-pulmonary monitors as we and others have done.[16,20,104] For example, we have used simultaneous, continuous measurement of temperature and HRIs in the NICU to find the body temperature where infants were most thermally stable.[104] Researchers have used HRV through continuous ECG recordings to measure autonomic maturation in VPT infants hospitalized in a NICU.[105] We recently completed enrollment of 352 infants (R01NR017872) employing HRIs[70] as an index of ANS dysfunction and predictor of infection.[62] In an interim analysis of 55 VPT infants, we found that more abnormal HRIs were associated with significantly increased odds of infection on that day.[103]

ANS Dysfunction Measured with Thermal Gradients and HRIs: Associations with Later ASD Symptoms. Preliminary findings[7] from a funded pilot study (PIs: Dail, Bradshaw) significantly contribute to the scientific basis for the proposed work. In this prospective, longitudinal trial of N=20 VPT infants (a subsample of participants enrolled in R01017872, PI: Dail), continuous measures of minute-by-minute thermal gradients and hour-by-hour abnormal heart rate indices, measured using Heart Rate Characteristic (HRC) scores derived from HeRO Monitors,[70] were collected for 28 days after birth (>40,000 samples/infant). Following NICU discharge, comprehensive neurodevelopmental follow-up consisting of standardized measures of cognition, language, and motor skills were collected at adjusted ages 6, 9, and 12 months for N=12 infants. At 12 months, assessments of social communication and early ASD symptoms were administered. Across all infants in the first 28 days of life, ANS dysfunction was indicated by an average HRC score >1 and abnormal negative thermal gradients for about 20% of the time (6 of 28 days). Neonatal abnormal HRC scores were strongly associated with ASD symptoms at 12 months ($r=0.81$, $p<.01$; **Fig. 3**), as was birth GA, birth weight (BW), and abnormal negative thermal gradients. A regression model including GA, BW, and HRC scores revealed a significant, unique predictive effect of abnormal HRI (HRC score) on ASD outcomes (model $r=0.86$, $p=.017$; HRC Score: $r=.72$, $p=.029$).[7] **This strong predictive model stands as compelling preliminary data for Aims 1-2.** This R01 is critical in delineating precise HRIs (e.g., tachycardia, bradycardia, HRV, etc.) and other indices (thermal gradients) that are associated with ASD outcomes. Each index may be sensitive to different underlying neural processes and parasympathetic-sympathetic dynamics. We will therefore go beyond the HRC scores to evaluate how different indices differentially map to ASD versus global ANS dysfunction.[106-109]

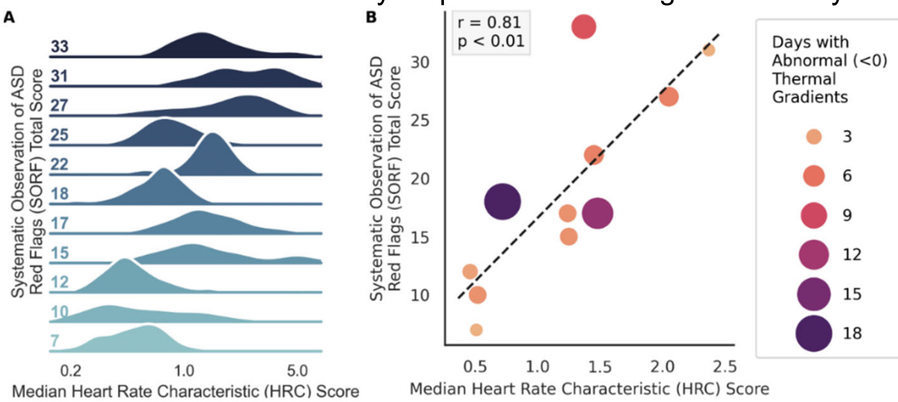


Figure 3. Association between neonatal abnormal heart rate characteristics (HRCs) and 1-year ASD symptoms. Higher scores indicate more heart rate abnormalities and more ASD symptoms. A) Ridgeline plot displaying the distribution of neonatal abnormal HRC scores across participant ASD symptoms (SORF total composite score) at 1 year. B) Scatterplot and associated regression between neonatal abnormal HRC scores across 1-year ASD symptoms (SORF total composite score).

Precise timing and dynamics of multiple ANS measures. HRC scores used in our preliminary data are coarse, hourly estimates of a single composite score. In this study, we will derive **more temporally resolved HRIs** from ECG recordings. Collecting temperatures continuously, with high temporal resolution, will allow for sophisticated analyses of the complex interplay of neonatal autonomic regulation across the preterm period.

Disentangling ASD from NDI in VPT Infants Across the First Three Years. Our preliminary data suggest that significant neurodevelopmental impairments did not emerge in VPTs until 12 months of age when over 50% of the sample exhibited social-communication delays and exceeded the clinical cutoff for ASD risk (Fig 4).[7] At 12 months, infants were characterized by ASD and NDI as follows: ASD features (high = 33%; moderate = 25%; little-to-none = 42%); NDI manifested by cognitive and/or motor delays (NDI = 67%; no NDI = 33%). 25% of the sample had no evidence of ASD or NDI. Of interest, 50% of the infants with high ASD features evidenced no motor or cognitive NDI. These preliminary descriptive statistics illustrate the **significant variability of developmental outcomes** for VPTs using direct clinical observation and ASD-specific measures. Minimal change in developmental skills was observed prior to 12 months, but significant developmental divergence was observed at 12 months. These **preliminary data** suggest **feasibility** for Aim 3: 1) participant retention and **completion of longitudinal follow-up measures** with VPT infants at evidence-informed time points (based on preliminary data); 2) obtention of **sufficient variability in developmental trajectories and ASD/NDI outcomes** with significant, distinct ASD and NDI symptoms already emerging at 12 months; and 3) use of measures with **sufficient sensitivity** to uniquely quantify ASD and NDI symptoms in infancy. Our data (Fig. 4) suggest significant variability and complexity in development and outcomes and Aim 3 is crucial for unveiling developmental nuances and advancing etiological frameworks of ASD in VPTs.

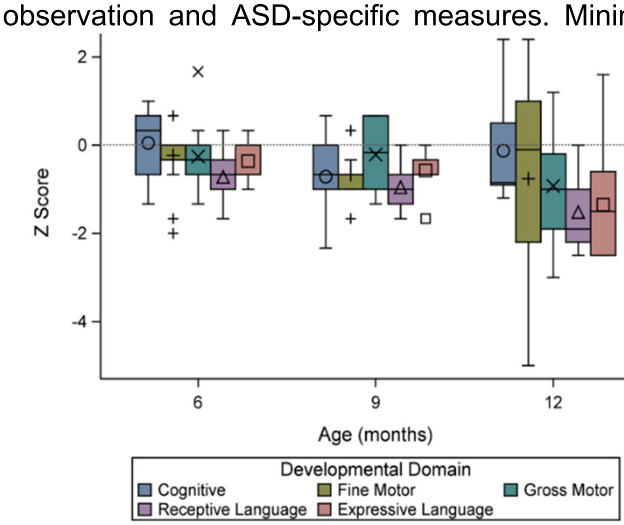


Figure 4. Bayley-4 subscale scores over time represented in z-scores where 0 is the average, -1 is one standard deviation (SD) below the mean, -2 is two SDs below the mean, etc.

Research Team

This study is a natural collaboration between **University of South Carolina's (USC)** MPI Robin B. (Knobel) Dail, Health Sciences Endowed Professor, College of Nursing (CON) and MPI Jessica Bradshaw, Associate Professor of Psychology, with the mutual interest of ANS dysfunction in infancy. The current project is an extension of the MPIs' individual research programs and will capitalize on existing infrastructure, resources, and ongoing/completed awards (Dail: R01NR017872;[62,82] Bradshaw: R01MH1364619, K23MH120476, foundation).

Neonatal Expertise: MPI Robin Dail is a nurse scientist with expertise in morbidity and mortality in PT infants. She has collaborated with **Prisma Richland NICU** neonatologists, on studies of ANS in VPTs over the last 6 years. **Co-I Ravi Rao**, Neonatologist and Medical Director will oversee a 50% research coordinator hired for the proposed study (see Budget Justification). **Co-I Kayla Everhart** is a CON neonatal researcher and has worked with MPI Dail since 2020. She is currently running a funded RCT in Prisma Richland's NICU with MPI Dail. Drs. Dail, Everhart, Rao, and research nurses/coordinators are all trained in collecting thermal gradient and

HRI data from Prisma NICU patients and maintaining a REDCap database with demographic and morbidity data. Dail's Data Manager will oversee data transfer/storage.

ASD, Neurodevelopment, and Longitudinal Data Expertise: MPI Jessica Bradshaw (Psychology) has expertise ASD/NDI in infants/toddlers and in prospective, longitudinal designs and longitudinal modeling.[28,85,110,111] Bradshaw maintains a lab staffed with 1 postdoc, 5 research coordinators, 1 psychologist, 2 recruitment coordinators, 5 graduate and 20 undergraduate students. Staff/students are trained in proposed data collection and processing procedures, 1 of whom will be dedicated to this project. 3 additional RAs will be hired specifically for this project (see *Budget Justification*) and Bradshaw's clinical staff (clinical psychology postdoc and social worker) will be available for RA training, case consultation, and family feedback when needed.

Physiological Data Science and Statistical Expertise: Co-I Christian O'Reilly (College of Engineering & Computing) is an electrical and biomedical engineer and neuroscientist with expertise in scientific computing, advanced statistical analyses, biosignal processing, and computational neuroscience with applications to neurodevelopment. He has established collaboration with the other investigators on this proposal (R01MH132925; Institutional grants).[7,108,112,113] Co-I O'Reilly will supervise 1 graduate student to support the analyses proposed in this project. **Co-I John Richards** (Psychology) is a renowned developmental cognitive neuroscientist with methods expertise in heart rate indices, EEG, and MRI as well as content expertise in nervous system development in PT infants.[17,114,115] Richards and Bradshaw have collaborated closely on studies of infant attention and physiology for the past 5 years (Co-I, R01MH132925; mentor K23MH120476) and has collaborated with Co-I O'Reilly on cortical source analyses in infancy.[113]

Study Design

This prospective, longitudinal study will enroll and follow 260 VPT infants. Minute-by-minute, continuous ANS monitoring will be provided during their entire NICU stay (admission duration range: 5-16 weeks). Following NICU discharge, comprehensive assessments will be administered at adjusted ages 12, 24, and 36 months.

Setting and Equipment. Participant enrollment will occur at Prisma Richland NICU, which is 2 miles away from USC campus where all study investigators are located. Standard-of-care electronic medical records (EMR), SpaceLabs cardiopulmonary monitors, and HeRO bedside monitors (equipped to store continuous R-R intervals, see *Letter of Support*), will be used for data acquisition. Central and peripheral temperatures will be recorded and saved using Grant Squirrel Dataloggers,[62] (n=26 available for use). This setting and available equipment allows for enrollment of 12 infants per month, with measures collected for an average of 3 months per infant. Prisma NICU capacity (70 beds) and admission rate (~250/yr) far exceeds this planned enrollment rate, ensuring study feasibility. **A NICU Research Nurse (RN) will dedicate 50% effort to this project** and will: obtain consent from mothers within 24 hours of the infant's birth, educate clinical staff on study protocols throughout the study, place and monitor infant instrumentation and skin condition (described below), monitor data acquisition, and download/enter/save all collected data (weekly for EMR data, 2x/month for physiological data). Upon NICU discharge, the RN will be responsible for ensuring all medical and demographic data from EMR is entered into a REDCap and all physiological data is transferred to USC Project Manager. All data will be housed at USC on a secure server. Developmental assessments (adjusted age 12, 24 and 36 months) will be completed in the participants' homes by MPI Bradshaw's research staff (team of 2 per visit). In-home data collection has resulted in adequate retention, particularly for our underrepresented populations, but families may come to the lab if they prefer. Research Specialists will have clinical experience and training in assessments to be administered. MPI Bradshaw's lab also has 5 clinical psychology graduate students, 4 of whom are supported by training grants (NIH T32, F31, NSF GRFP), and who can contribute to clinical data collection and training as needed. Our team has conducted in-home data collection for >5 years and has designed custom hardware and software solutions. Our in-home data collection kits include 5 A/V recording systems, assessment kits (2 Bayley-4s, 3 CSBSs, 3 ADOSs), and 3 Surface Pros for completing REDCap surveys. Due to different home sizes and resources, we bring a variety of infant highchairs, adult chairs, and tables to ensure a standardized setup in any context. Data will be entered into REDCap and videos stored on secure servers.

Participants. Over the first two years of the study, we will enroll a total of 260 VPT infants born at Prisma Richland NICU between **24 and 31 6/7 weeks GA and at least 500 grams birthweight.** Infants born less than 24 weeks GA are fragile and frequently unstable. We consistently have used these enrollment GAs and weights for our studies with little attrition due to instability or mortality.[61,62,116] <u>Exclusion criteria</u> include: major anatomical or cardiac defect known at birth, anatomical brain damage or known in utero infant asphyxia, and known genetic or chromosomal abnormality that is known to affect heart function or neurodevelopment (e.g., Down Syndrome, Fragile X). <u>To account for attrition</u>, we are over-enrolling by 30%. Infants may be withdrawn from the study by parents or attending physician during NICU hospitalization, or a very few may die due to complications of prematurity. Based on our pilot data and previous studies, enrolling 260 VPT infants in the NICU should yield at least 200 retained infants who will complete the final 3-year assessment visit and enable sufficiently powered analyses for our aims. We expect to enroll n=12/month. Prisma NICU admits 200-250 VPTs annually (~50% males; 56% Black/African American, 35% White, 9% Hispanic). We will monitor enrollment demographics to ensure our sample is representative of NICU hospital admission and our analyses will account for sex as a biological variable. Given the NICU capacity and admissions rate, our enrollment success rate (90% for NICU families), and our conservative plan to over-enroll by 30%, we will easily reach our target enrollment within 2yrs.

Procedures

All study procedures are outlined in PHS Human Subjects and Clinical Trials Information form. Consent will be obtained from parents before their infant is born or after birth up to 24 hours of age. Data will be collected on the primary variables of abdominal and foot skin temperature, and ECG for each infant included in the study until NICU discharge, transfer, or death. Demographic data, maternal/prenatal history, family history of ASD, and PT-related morbidity data will be collected from the EMR and follow-up questionnaires to be used as covariates (race, maternal education, GA) and moderators (sex, morbidities) in analyses. Family history of ASD will be assessed via parent interview and ASD screeners for any biological siblings. Developmental assessments will begin at 12 months, following a virtual touch point at 6 months with parents. This virtual touch point will take place over the phone and will include a brief medical history since NICU discharge, re-explanation of the study and study procedures, demographic update (contact information, maternal education, income), and visit scheduling. Follow-up data will be collected in participant homes at adjusted ages 12, 24 and 36 months.

Primary Variables/Measures

Neonatal ANS Function: Thermal Gradients. Abdominal (central) and foot (peripheral) temperatures will be measured continuously with a thermistor (disposable skin sensor, 499B, Cincinnati Sub Zero, Cincinnati, OH) covered by Mepitac Silicone adhesive tape.[62] These thermistors and tape are used for temperature monitoring in standard care. Thermistors record temperatures every minute using portable, high-precision dataloggers (Squirrel SQ data logger, Grant Instruments), resulting in a CPTd value every minute, as done in our previous studies.[61,62] Abnormal thermal gradients are either negative (CPTd < 0º C; foot temperature warmer than abdomen) or high (CpTd > 2º C; abdominal temperature much warmer than foot). Per each 24-hour period, days with < 80% of valid temperature data will be dropped. Our previous studies suggest this method results in < 10% data loss. **Heart Rate Indices: Pre-Processing.** HRIs will be derived from continuous ECG recordings from standard of care cardiopulmonary and HeRO monitors within the NICU (see *Letter of Support*). Continuous, beat-by-beat R-R intervals will be recorded from birth to discharge via the FDA-approved device HeRO Solo from Medical Predictive Science Corporation (MPSC). This same system was used for preliminary data reported here.[7] MPSC utilizes an FDA-approved, automated QRS detector to remove noise/artifacts and to detect R-waves; it has been validated against hand-annotated ECG waveforms with precision >99% and sensitivity >90%.[117,118] Further validation is evident in its use in the largest randomized controlled trial conducted among VPTs, where displaying HRIs based on R-R intervals generated by the MPSC QRS detector to clinicians resulted in a 20% reduction in all-cause mortality and a 40% reduction in mortality after infection.[119,120] In addition, our team has developed custom pre-processing pipelines that will further validate the MPSC-generated R-R intervals, as we have done in previous work.[108,110,121] Specifically, isolated missed beats will be detected using standard non-parametric outlier analysis and corrected through interpolation. Segments with >5 contiguous missing/edited beats will be rejected from analysis. Windows of 1h with >10% discarded data will be inspected and dropped as needed. This procedure will be validated and adjusted using a large dataset of manually cleaned ECG available from previous and ongoing projects run in Bradshaw's lab. **HRIs**. The following HRIs will be extracted: HRV, RSA, NN sample entropy, mean NN, NN standard deviation, and NN skewness.[67] To support comparison, we will also include four ECG-derived features that we used in a previous study,[108] namely, the root mean square of the difference between successive NN (RMSSD), the ratio of the two main axis of the ellipse fitting the scatter plot of NN vs preceding NN (i.e., Poincaré plot; SD1SD2), the coefficient of variation (std/mean) of NN (CVNN), and the HRV triangular index (HTI) calculated as the number of NN values in the modal bin of the normalized 8-ms-bin NN histogram. Continuous wavelet transforms (CWT) will be used to estimate RSA (see Table 2b below). We will complement our analysis of the relationship between ASD and ANS by refining our assessment of sympathetic and parasympathetic branches of this system. Toward this goal, we will investigate the relationship between ASD and the Sympathetic and Parasympathetic Activity Indices (SAI and PAI),[106] the Cardiac Sympathetic Index (CSI), and the Cardiovagal Index (CVI).[107,108] This extensive set of indices will be used specifically for ML prediction of ASD and NDI (Aim 1).

PT-associated Morbidities (birth to NICU discharge). Analyses for Aim 2 will use an established Morbidity Score previously adopted to study associations between ANS development and PT-related morbidities in the NICU.[20] This summative morbidity score quantifies the number of morbidities experienced at any time and includes the following factors: incidence of culture positive infections, NEC diagnosis, presence of PDA requiring medical or surgical intervention, severe brain injury (ultrasound readings of Grade III or IV intraventricular hemorrhage and/or periventricular leukomalacia), diagnosis of BPD or chronic lung disease, need for mechanical ventilation, and ROP at ≥ stage I (Table 1).[20] Morbidities will be extracted from the EMR weekly.

Social Drivers of Health (SDOH) and Early Infant Experience. Given the significant effect of SDOH, and the prenatal and postnatal environment on infant/maternal care and the developing brain,[122,123] the following measures will be accounted for in analyses. At consent (birth) and outcome (3 years), mothers will complete the 15-item, evidence-based, patient-centered tool PRAPARE (Protocol for Responding to and Assessing Patients' Assets, Risks, and Experiences).[87,88] PRAPARE measures SDOH across four domains: 1) personal characteristics (e.g., race, ethnicity), 2) family and home, 3) money and resources, and 4) social-emotional .health, and provides an overall Risk Tally Score that represents the number of social drivers of health risk (range: 0-22). PRAPARE is a recommended for assessing maternal unmet social needs as potential barriers to healthy prenatal and postnatal outcomes.[124] In addition, the following four measures (variables) will be collected on an ongoing basis to evaluate infant caregiving context and early life experience[125]: maternal perceived stress (any moderate to high score > 20),[126,127] depression (any score > 12)[128,129], infant health (number of days of infant post-discharge hospitalization), and infant developmental support (months of early intervention participation). Note that clinically significant/elevated levels of maternal stress/depression, as defined by the instruments, will be flagged immediately and caregiver will be contacted by the lab's clinical psychologist and/or MPI Bradshaw (licensed clinical psychologist) who will provide resources/referrals as needed (same protocol as ongoing R01).

ASD and NDI Outcome (36 Months): ASD Symptoms. The ADOS-2 will be administered at age 36 months adjusted and Calibrated Severity Scores (CSS)[130] for Social Affect (SA CSS) and Restricted and Repetitive Behavior (RRB CSS) will be used as dimensional measures of ASD outcomes in Aim 1-2. ADOSs will be administered by clinicians trained to research reliability and supervised by MPI Bradshaw and lab psychologist, with co-scoring to consensus occurring for at least 20% of the sample. CSS ranges from 1-10 and represents an excellent measure of ASD symptoms, relatively independent from IQ and language abilities,[130,131] that has been shown by our group and others to have sufficient variability for biomarker prediction.[132,133] For Aim 3, the Total CSS score will be used to compare derived clusters. **Neurodevelopmental Impairment**. Consensus is lacking on the definition of severe NDI outcome for PT populations.[134] But when assessing NDI, studies tend to use normative thresholds to define impairment (e.g., 1.5 or 2 standard deviations below the mean) for the following areas: motor, cognitive, language.[15,134] However, this dichotomous approach fails to capture subtle challenges and nuanced profiles in VPT infants.[135] This study will use a dimensional measure of neurodevelopmental impairment as the NDI outcome for Aims 1-2. In line with other studies of neurodevelopmental outcome in VPTs,[136] the Bayley Scales of Infant and Toddler Development, 4th Edition (Bayley-4)[137] will be administered at adjusted age 36 months, and the Cognitive, Motor, and Language standard scores (mean=100, SD=15) will be used as continuous measures of impairment. These scales explain a sizable proportion of later cognitive and academic outcome variability and are invaluable for characterizing early development and guiding intervention efforts.[138]

Emerging ASD and NDI (12-24 Months): ASD Symptomatology. Emerging ASD symptom profiles will use a direct measure of very early ASD symptoms, as has been used in our previous studies,[7,18,28,29,52,85] at 12 and 24 months. Specifically, we will use the Systematic Observation Rating of Flags for ASD (SORF), an observational measure of ASD red flags based on the DSM-5 framework that will be scored from the Communication and Symbolic Behavior Scales-Behavior Sample (CSBS).[139] The SORF has been found to distinguish children with ASD from children with NDI (but no ASD) and typically developing children, and our study team has documented the sensitivity and specificity of this measure for differentiating 12-month-old infants later diagnosed with and without ASD.[140] In infants 16-24 months of age, the SORF has large effects and good discrimination (area under the curve=.84-.87).[140] For Aim 3, the SORF Social Communication Domain score and the SORF RRB Domain score will be used to characterize emerging ASD symptoms. **Neurodevelopmental Impairment.** NDI will be measured dimensionally using standard scores from all 5 subscales of the Bayley-4: Cognitive, Fine Motor, Gross Motor, Receptive Communication, and Expressive Language. Scores will be adjusted for infants' GA. These subscales will be used in cluster analyses described in Aim 3.

Data Management

A HIPAA-compliant, USC-based, REDCap database will be created to collect all data in this study. The NICU RN will review and enter neonatal demographic and clinical data. Direct import from EMR into REDCap is not feasible as an RN is required to harmonize EMR data across individual patients to be suitable for analysis. Per our recent 5-site R01, this protocol involves clinical reading and interpretation of multiple tests results with varying response options. Co-I Everhart will train the RN to reliability in this process and will oversee all data harmonization procedures, as she has done for over 350 VPT infants in our multisite R01. The RN will download and store temperature data from dataloggers and ECG data from HeRO monitors every two weeks until the infant is discharged from the NICU. All data will be accessible over an encrypted and password-protected connection. The Data Manager at USC will clean temperature data and transfer data into SAS (9.4) datasets. Co-I O'Reilly's team will pre-process ECG data and compute HRIs (see above). Clinical data collected after NICU discharge will be stored on our secure server (videos) and entered in REDCap (assessment scores) following each study visit. Double data entry will be used to ensure data validity.

Data Analysis

Aim 1: Examine PT ANS dysfunction as a specific predictor of ASD symptoms at outcome. The analysis for Aim 1 will leverage ML by using a **support vector regression** from Scikit-Learn[141] to implement non-linear regression models that predict symptoms of ASD (Social and RRB CSS from ADOS-2) and NDI (Cognitive, Motor, Language standard scores from Bayley-4) at outcome from a set of features describing the preterm ANS and its dysfunction. **Table 2a** lists the measures that will be considered for this analysis. For each measure, four primary features will be extracted: 1st, 50th, and 99th percentiles, and the inter-quartile range for the measured values across the NICU stay. The slope of the relationship between the median value of these measures (averaged per 24-h period) and GA will also be considered. We will also investigate cross-modal temporal associations by calculating the correlation across time between all pairs of median measures (elevated per 1h-hour period) and Fisher Z-Transforming these correlations. We will add these correlations to the feature set if their intra/interparticipant variance ratio (estimated with bootstrapping) is statistically significant compared to chance-level assessed on a surrogate dataset obtained by randomizing the temporal order of the median measure time series. The correlation matrix between these features will be hierarchically clustered using the Seaborn *clustermap* function, and highly correlated features will be pruned to remove multicollinearity. Linearity of this relationship will be tested, and a power transform used if necessary. We will further evaluate the association of these features with demographic covariates (race, sex, maternal education, GA) and, if statistically significant, the effect of these covariates will be regressed out of the features.

Table 2a. ANS Measure Descriptions for Aim 1 Machine Learning

Measures	Modality	Interpretation
% time CPTd < 0^{o} C	Temperature	Peripheral body warmer than core body, indicative of ANS dysfunction or immaturity[59,123]
% time CPTd > 2^{o} C	Temperature	High core to peripheral temperature indicative of Sympathetic activity, seen with sepsis and stressor activity in PT infants[89,124]
SAI	Heart Rate	Index of sympathetic activity
PAI	Heart Rate	Index of parasympathetic activity
CSI	Heart Rate	Index of sympathetic activity
CVI	Heart Rate	Index of parasympathetic activity
HRV	Heart Rate	Measured as the natural logarithm of RMSSD. Decreased values indicate ANS dysfunction and are characteristic among individuals with ASD[125]
Mean NN	Heart Rate	Increased with overactive sympathetic and decreased parasympathetic or vagal tone.[16] Highly predictive features for likelihood of ASD according to previous study.[99]
NN standard deviation	Heart Rate	Spread of the NN distribution. Index both sympathetic and parasympathetic activity.[126]
NN skewness	Heart Rate	Transient HR decelerations (bradycardia) result in negative skewness (parasympathetic). Transient HR accelerations (tachycardia) result in positive skewness (sympathetic).
NN sample entropy	Heart Rate	Decreased values are thought to be associated with a decoupling of the ANS from external inputs (i.e., non-responsiveness, non-adaptability).[127,128]
RMSSD	Heart Rate	Parasympathetic activity; vagally mediated variation in heart rate.[129]
SD1SD2	Heart Rate	Reflect the balance between short-term vagal activity and sympathetic modulation.[129]
CVNN	Heart Rate	STD(NN) captures both low and high-frequency components of the heart-rate signals. High-frequencies variations are due to RSA (parasympathetic). The normalization of STD(NN) by MEAN(NN) makes CVNN more stable when comparing individuals or conditions with different baselines.[129]
HTI	Heart rate	Depressed HTI is a robust index of sympathovagal imbalance.[130]
RSA	Respiration & Heart Rate	Periodic slowing and speeding up of heart rate with the respiratory cycle. Higher RSA indexes more vagal flexibility, which promotes balanced parasympathetic and sympathetic influences to the heart.

Table 2b. RSA Calculation

RSA Calculation Method: Continuous Wavelet Transform (CWT)	
Step	**Relevant References**
1. Transform R-R (beat-by-beat) series into a sample-rate (10Hz) time series, then use "cwt" function of MATLAB. The output from cwt is all the **wavelet-transformed IBI series,** that is the IBI time series transformed by the wavelet filterbank.	*Parameters used*: "analytic Morlet", "amor", and 48 "voices per octave" (# of wavelets). *CWT Analysis and RSA*: Grossmann & Morlet, 1984; Cnockaert et al., 2008; Wachowiak et al., 2016
2. Use inverse CWT ("icwt") with frequency bands set to the min and max of respiration, determined by age-appropriate norms. The output of icwt is the **RSA time series** for the respiration frequency chosen for this age.	*Respiration rate in infants*: Gagliardi & Rusconi, 1997; O'Leary et al., 2015; Rusconi et al., 1994; Tveiten et al., 2016
3. To be consistent with established RSA calculation methods (PB-MPF), use log(var(icwt time series)) to **calculate RSA.**	*Porges-Bohrer Moving Polynomial Filter*: Porges, 1982; Porges & Bohrer, 1990
Advantages. Compared to other filters, the CWT approach results in higher resolution for frequencies, better drop-off, and is more modifiable.	

Social Drivers of Health (SDOH) and Early Infant Experience. Further, we will test for significant associations between these features and relevant SDOH (PRAPARE Risk Tally Score), and early infant experience (maternal stress, maternal depression, infant health, infant intervention) variables. If these associations are statistically significant, the effect of these covariates on the features will be regressed out. We recognize the complex, interconnected, and multifaceted nature of demographics, SDOH, and early infant experience, all of which cannot be fully and meaningfully captured in a single score, while also acknowledging the possibly high number of influential variables and the collinearity therein, which raises significant analytic challenges.[142] Thus, our analytic plan for Aims 1-2 will account for these summative variables, while also allowing us to further probe individual items in subsequent analyses.

We will then use permutation feature importance techniques developed in the context of explainable artificial intelligence and implemented in Scikit-Learn[141] to identify which features contributed the most to this prediction. Since most of our features can be interpreted in a mechanistic way (e.g., see Table 2a), this analysis will allow us to confirm and refine our hypotheses on the role of ANS dysfunction in the emergence of ASD and NDI. All ML analyses will be conducted using nested subject-level-cross-validation (train: 70%; test: 15%; validation: 15%) to allow reliable and reproducible model fitting and hyperparameter tuning. The effect size for ANS as a predictor of outcomes at 3 years of age will be evaluated as model's R^2 values (i.e., coefficients of determination) and their statistical significance will be assessed by looking at whether their bootstrapped 95% confidence intervals overlap with 0. Since model assessment is cross-validated, negative R^2 values and confidence intervals overlapping with 0 are expected in the absence of a true relationship.

Aim 2: Evaluate preterm morbidities and sex as potential moderators in the association between PT ANS dysfunction and symptoms of ASD and NDI at outcome. While Aim 1 (above) leverages machine learning, the statistical analysis for this Aim 2 relies on a **classical moderation modeling** where the association between PT ANS dysfunction (independent variables; listed in the next paragraph) and symptoms of ASD and NDI at outcome (dependent variables) will be tested for **moderation by sex and a previously validated preterm morbidity index,**[20] as defined in the PT Morbidities section. The linearity of this relationship will be tested, and a power transform will be used if necessary. The correlation matrix between independent features will be analyzed and highly correlated features will be discarded to avoid multicollinearity issues. If a significant moderation effect is found for the morbidity index, we will conduct post hoc moderation analyses to determine which of its constituting factors (severe brain injury,[143] infection, or NEC[145], BPD[75,146] or need for mechanical ventilation,[147] ROP,[149] and/or PDA) contribute to the moderation. Python (StatsModels Mediation class), which supports both mediation and moderation analyses, will be used.

We will use the same dependent variables as those used in Aim 1 (i.e., ADOS-2 CSS and Bayley-4 Scales). Our analyses will align with our preliminary investigations[7] and will characterize ANS function during PT periods using 1) the proportion of days during which negative thermal gradients are present > 30% of the time, 2) the proportion of days during which CPTd > 2º C are present > 10% of the time, 3) IBI skewness (based on its key role in HRI predictors used in our previous work[53] and the work of others[67]), and 4) a small set ($\leq$ 3) of features that will be shown to have the highest predictive power in Aim 1. By using a combination of ANS measures that are determined based on preliminary data and that are highly predictive according to the analyses of Aim 1, we ensure that Aim 2 does not significantly depend on Aim 1. We will correct for multiple independent tests using the Bonferroni method. We will first perform these analyses using average values (e.g., rates and frequencies of ANS measures; no repeated measures). Then, we will add the effect of GA by considering 24-h averages for the independent variables and replace the fixed effect linear moderation by a mixed-effect one, including both the intercept and slope of the effect of GA. For this second analysis, we will test if GA needs to be transformed (e.g., power transform) prior to its use for linear modeling and evaluate the effects of potential covariates, including SDOH and early infant experience, as listed above.

Aim 3: Characterize and differentiate the emergence of ASD symptoms and NDI in VPT infants. We will characterize the development of VPT infants using direct measures of ASD symptoms and NDI administered at ages 12 and 24 months. Given the developmental heterogeneity observed in both infants with ASD and VPT infants, we will use both SORF subdomains (Social Communication and RRB Domain scores) to characterize ASD symptoms and all subtests of the Bayley-4 to characterize NDI (Cognitive, Fine Motor, Gross Motor, Receptive Communication, Expressive Communication). We will use a supervised clustering approach to identify subgroups of developmental trajectories that differ in terms of the emergence of ASD symptoms and NDI and that are predictive of the outcome at 36 months. To better capture the trajectories of autonomic ASD characteristics, and considering that we have two time points, we will consider the mean value across time points and the slope between time points for the two ASD and five NDI subscales, for a total of 2 x 7=14 features. The supervised approach will compute SHAP (Shapely Additive exPlanation) values associated with the features used for an XGBoost regression predicting the ADOS CSS Total score at 36 months (5-fold cross validation).[150] The SHAP values will then be used as input to the unsupervised The DBSCAN algorithm for clustering.[151,152] This approach will be implemented using Scikit -Learn[141] and the SHAP Python package.[153] Although the nonlinear XGBOOST and DBSCAN algorithms are not vulnerable to multicollinearity issues, we will use a Principal Component Analysis to evaluate the proportion of variance captured by every component and reduce redundancy in the feature set, if necessary. The Silhouette Coefficient[154] will be computed to assess the qualify of the clustering, the t-distributed Stochastic Neighbor Embedding approach[155] will be used to visualize the separability of the clusters, and the stability of the clustering will be assessed through bootstrapping.[156,157]

By using a supervised clustering approach based on the outcome, we maximize our chances of obtaining stable clusters that group trajectories predictive of different outcomes. These clusters will, therefore, support the possibility of using early symptom trajectories to predict ASD diagnosis. Clusters will be visualized as seven subplots (one per ASD and NDI subscales) with the time points along the x-axis and the cluster mean values in y-axis with bootstrapped 95% confidence intervals as obtained from Python Seaborn *lineplot* function. We will test whether extracted clusters form statistically significant subgroups by testing differences in centroids using a Hotelling T-squared test. We will validate the clinical significance of extracted clusters by assessing whether subgroups differ based on 36-month ASD outcome (ADOS CSS) using t-tests, adjusted for multiple comparisons.

Sample Size and Statistical Power Considerations. Aim 1 uses a ML-based regression approach to establish the statistical and clinical significance of the relationship between PT ANS abnormality and ASD/NDI outcomes. For this aim, the R^2 values for regression model fitting can be seen as the squared values of Pearson's coefficient of correlation which can be used for power analysis. With N=200, we can expect to detect a correlation of at least 0.20 with a 20% false negative rate (i.e., β=0.2) and a 5% false positive rate (i.e., α=0.05). This coefficient of correlation corresponds to a 0.04 R^2 value, which can be interpreted as the percentage of variance explained. Once corrected for 7 independent tests (i.e., 7 subscales, Bonferroni correction), this R^2 increases to about 0.0625. Therefore, we consider this sample size sufficient, as effects that cannot explain reliably more than 6.25% of the total variance in the final diagnosis are probably of low clinical significance. For **Aim 2**, we used G*Power[158] to determine the minimal sample size required to have a 0.80 power (i.e., β=0.2) to test for moderation (two moderators: medical morbidity index and sex) in a model including six predictor main effects and their centered interaction with moderators, and using an α=0.01 (i.e., 0.05 Bonferroni-corrected for testing 5 dependent variables). For a f^2=0.25 (considered a medium effect size), we will need N=130, which demonstrates that we have a large enough sample to run our main analysis and expect to obtain significant results if a clinically relevant effect exists. Our larger sample size (N=200) affords us extra maneuvering margin in case of smaller effects and for integrating co-variates (race, GA, and maternal education). For **Aim 3**, we will be able to detect significant differences in multivariate means between pairs of clusters each grouping 50 participants with β=0.2, α=0.05, and 14 response variables if the effect size has at least a Mahalanobis distance (a multivariate generalization of the Cohen's d) of 0.92, implying that we will be able to establish that these clusters are different only for relatively large effect sizes. If significant effects cannot be detected with our sample, we will increase the statistical power of this test by reducing the number of clusters (hence increasing the number of participants per cluster) and reducing the number of response variables (i.e., feature selection). Similarly, we will be able to detect effects having a Cohen's d of at least 0.57 for differences in mean ASD symptoms (Total CSS) between pairs of clusters with N=50 each (power=0.8; α=0.05; two-tailed).

Potential Challenges and Alternative Solutions. Many of the potential problems and alternative solutions have been uncovered in our previous studies, and effective solutions have always been implemented. As in the past, all potential issues will be discussed with study team members and immediately addressed as needed and as outlined in the MPI plan. All-investigator meetings will occur biannually to review progress, identify challenges, and generate solutions. **Enrollment**: Due to the ebb and flow of Prisma Richland NICU enrollment, we have adopted conservative enrollment quotas with a minimum of 12 infants per month (~50% of their annual VPT admissions), which allows us to over-enroll by 30% to account for attrition and loss to follow-up. For further backup, we have another NICU in the Prisma system in Greenville, SC that we can use as a second site for enrollment. Thus, we are confident we will meet our enrollment quotas for this study. **Follow-up Assessments**. We have built in a 6-month virtual visit (and incentive) to retain infants from discharge to the first follow-up ~1 year later. All follow-up assessments will be completed in the families' homes at their convenience to increase research accessibility. This strategy has greatly increased participation in our current studies, but in-lab visits are still available if desired. We will also access our Community Advisory Board as needed to overcome any barriers. All neonatal morbidities and severity will be recorded and could be used for Aim 2 in case the morbidity score shows limited variability. Later morbidities that may impede diagnostic determination (e.g., severe motor delays) will be evaluated in all-investigator meetings to determine course of action for inclusion in analyses.